Importing the buddy move data set

```
import pandas as pd
Df = pd.read_csv("/buddymove_holidayiq.csv")
Df
```

	User Id	Sports	Religious	Nature	Theatre	Shopping	Picnic
0	User 1	2	77	79	69	68	95
1	User 2	2	62	76	76	69	68
2	User 3	2	50	97	87	50	75
3	User 4	2	68	77	95	76	61
4	User 5	2	98	54	59	95	86
...	...	...	...	...	...	...	...
244	User 245	18	139	148	129	129	168
245	User 246	22	114	228	104	84	168
246	User 247	20	124	178	104	158	174
247	User 248	20	133	149	139	144	213
248	User 249	20	143	149	139	159	143

249 rows × 7 columns

1. Head(n) function- It is used to display the n rows from the beginning of the data set

```
Df.head(20)
```

	User Id	Sports	Religious	Nature	Theatre	Shopping	Picnic
0	User 1	2	77	79	69	68	95
1	User 2	2	62	76	76	69	68
2	User 3	2	50	97	87	50	75
3	User 4	2	68	77	95	76	61
4	User 5	2	98	54	59	95	86
5	User 6	3	52	109	93	52	76
6	User 7	3	64	85	82	73	69
7	User 8	3	54	107	92	54	76
8	User 9	3	64	108	64	54	93
9	User 10	3	86	76	74	74	103

2. The shape attribute of pandas- DataFrame stores the number of rows and columns as a tuple (number of rows, number of columns)

12	User 13	3	64	82	82	75	60

```
1 shape = Df.shape
2 shape
```

```
(249, 7)
```

19	User 20	5	114	83	65	114	102

3. Info of the dataset- The info() method prints information about the DataFrame. The information contains the number of columns, column labels, column data types, memory usage, range index, and the number of cells in each column (non-null values)

```
1 Df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 249 entries, 0 to 248
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype 
---  ------     --------------  ----- 
 0   User Id    249 non-null    object
 1   Sports     249 non-null    int64 
 2   Religious  249 non-null    int64 
 3   Nature     249 non-null    int64 
 4   Theatre    249 non-null    int64 
 5   Shopping   249 non-null    int64 
 6   Picnic     249 non-null    int64 
dtypes: int64(6), object(1)
memory usage: 13.7+ KB
```

4. Summary of the dataset- The describe() function computes a summary of statistics pertaining to the DataFrame columns. This function gives the mean, std and IQR values.

And, function excludes the character columns and given summary about numeric columns.

```
1 Df.describe()
```

	Sports	Religious	Nature	Theatre	Shopping	Picnic
count	249.000000	249.000000	249.000000	249.000000	249.000000	249.000000
mean	11.987952	109.779116	124.518072	116.377510	112.638554	120.401606
std	6.616501	32.454115	45.639372	32.132696	41.562888	32.633339
min	2.000000	50.000000	52.000000	59.000000	50.000000	61.000000
25%	6.000000	84.000000	89.000000	93.000000	79.000000	92.000000
50%	12.000000	104.000000	119.000000	113.000000	104.000000	119.000000
75%	18.000000	132.000000	153.000000	138.000000	138.000000	143.000000
max	25.000000	203.000000	318.000000	213.000000	233.000000	218.000000

5. Handling missing value- In order to check missing values in Pandas DataFrame, we use a function isnull() and notnull(). Both function help in checking whether a value is NaN or not. These function can also be used in Pandas Series in order to find null values in a series.

```
1 Df.isnull()
```

	User Id	Sports	Religious	Nature	Theatre	Shopping	Picnic
2	False	False	False	False	False	False	False

There are no null values as all columns are false

6. Handling duplicate records

```
Df.drop_duplicates(subset ="User Id",
                   keep = False, inplace = True)
Df
```

	User Id	Sports	Religious	Nature	Theatre	Shopping	Picnic
0	User 1	2	77	79	69	68	95
2	User 3	2	50	97	87	50	75
3	User 4	2	68	77	95	76	61

There are duplicate value present in values other than user id, which need not be removed here as they are relevent.

7. Handling outlier- An Outlier is a data-item/object that deviates significantly from the rest of the (so-called normal)objects. They can be caused by measurement or execution errors.

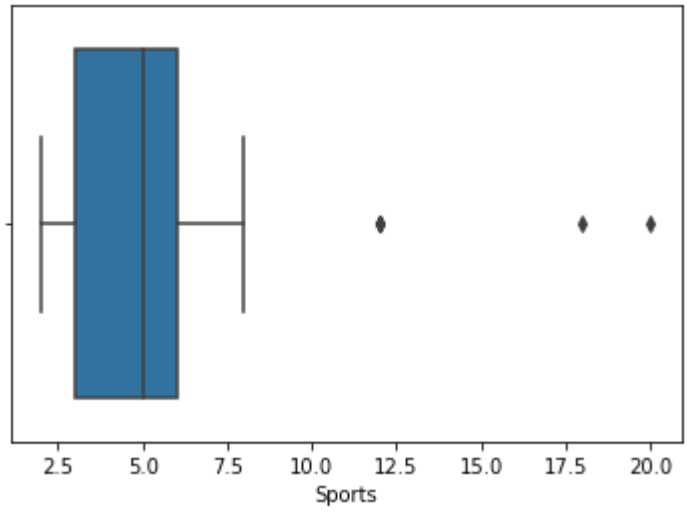
Visualization- Box Plot

It captures the summary of the data effectively and efficiently with only a simple box and whiskers. Boxplot summarizes sample data using 25th, 50th, and 75th percentiles. One can just get insights(quartiles, median, and outliers) into the dataset by just looking at its boxplot.

30	User 31	4	51	115	110	51	84

```
1 import seaborn as sns
2 sns.boxplot(Df['Sports'])
```

```
/usr/local/lib/python3.7/dist-packages/seaborn/_decorators.py:43: FutureWarning: Pass
  FutureWarning
<matplotlib.axes._subplots.AxesSubplot at 0x7f91cd4cacd0>
```



*Outliers may be present after 10.0. *

```
1 import numpy as np
2 print(np.where(Df['Sports']>10.0))
```

```
(array([23, 24, 25, 26, 27]),)
```

```
1 Df.head()
```

	User Id	Sports	Religious	Nature	Theatre	Shopping	Picnic
0	User 1	2	77	79	69	68	95
2	User 3	2	50	97	87	50	75
3	User 4	2	68	77	95	76	61
4	User 5	2	98	54	59	95	86
5	User 6	3	52	109	93	52	76

8. Variable identification: define each variable and its role in the dataset

```
1 type(Df)
```

```
pandas.core.frame.DataFrame
```

```
1   Df['Shopping'].dtype
```

```
dtype('int64')
```

```
1 Df['User Id'].dtype
```

```
dtype('O')
```

```
1 Df['Religious'].dtype
```

```
dtype('int64')
```

```
1 Df['Sports'].dtype
```

```
dtype('int64')
```

```
1 Df.dtypes
```

```
User Id      object
Sports        int64
Religious     int64
Nature        int64
Theatre       int64
Shopping      int64
Picnic        int64
dtype: object
```

We can see the data types of all columns/attributes

9. Univariate analysis: for continuous variables, build box plots or histograms for each variable independently

```
1 Df['Sports'].mean()
```

```
6.071428571428571
```

```
1 Df['Sports'].median()
```

```
5.0
```

```
1 Df['Sports'].std()
```

```
4.626242146939177
```

```
1 Df['Sports'].value_counts()
```

```
5     6
4     5
2     4
3     4
6     3
12    3
8     1
18    1
20    1
Name: Sports, dtype: int64
```

BOX PLOTS

```
1 import matplotlib.pyplot as plt
2
3 Df.boxplot(column=['Sports'], grid=False, color='black')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d26e5f10>
```




```
1 import matplotlib.pyplot as plt
2
3 Df.boxplot(column=['Religious'], grid=False, color='black')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d0541750>
```




```
1 import matplotlib.pyplot as plt
2
3 Df.boxplot(column=['Shopping'], grid=False, color='black')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d04cd090>
```



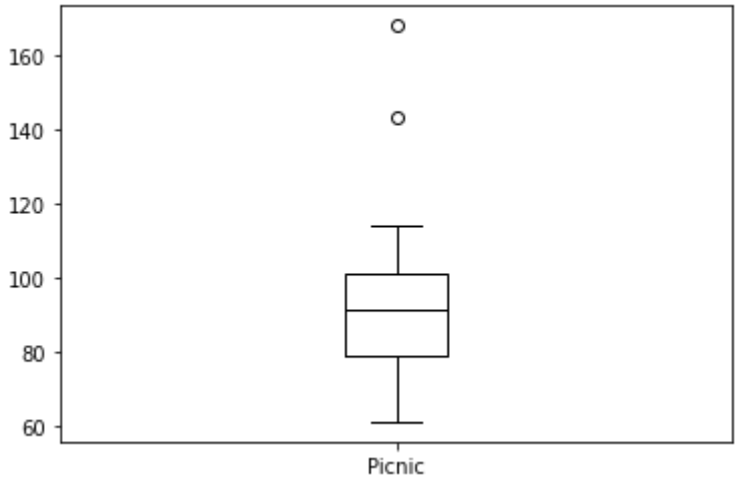

```
1 import matplotlib.pyplot as plt
2
3 Df.boxplot(column=['Nature'], grid=False, color='black')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d04b1450>
```



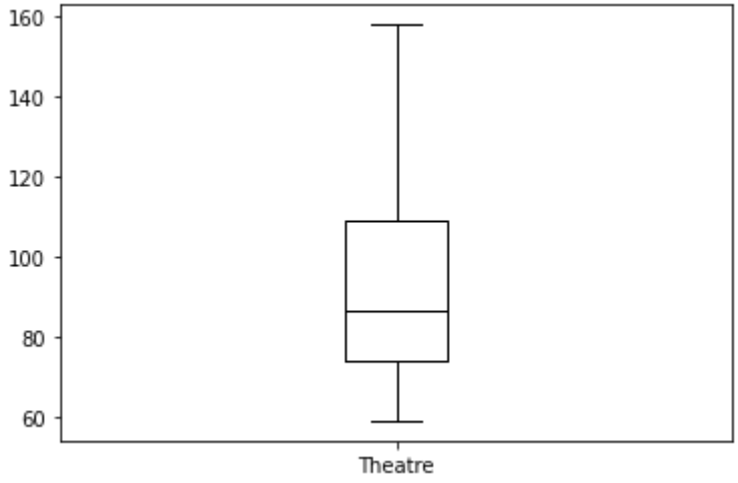
```
1  import matplotlib.pyplot as plt
2
3  Df.boxplot(column=['Picnic'], grid=False, color='black')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d0297f90>
```




```
1 import matplotlib.pyplot as plt
2
3 Df.boxplot(column=['Theatre'], grid=False, color='black')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d029e410>
```



HISTOGRAM

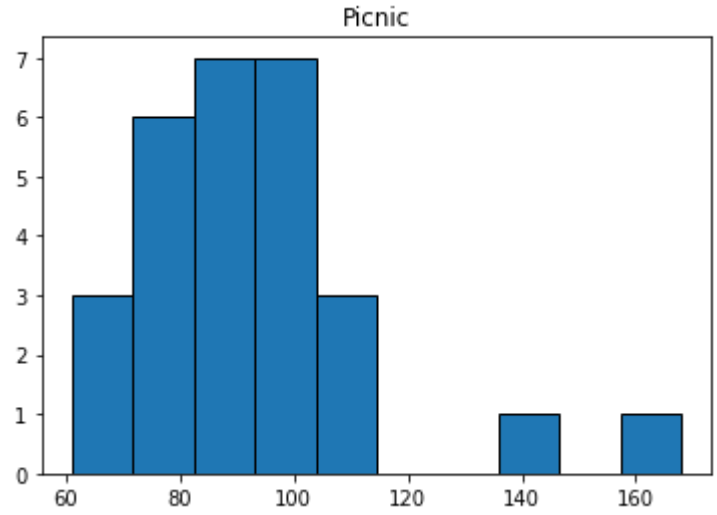
```
1 import matplotlib.pyplot as plt
2
3 Df.hist(column='Sports', grid=False, edgecolor='black')
```

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f91d037a8d0>]],
      dtype=object)
```



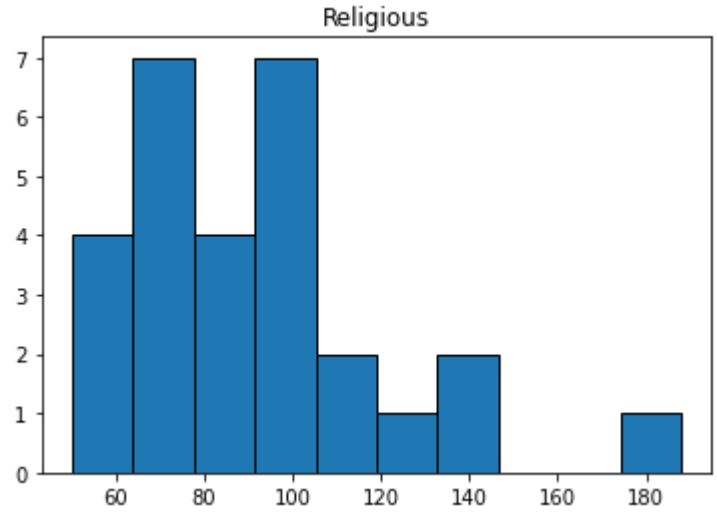

```
1 import matplotlib.pyplot as plt
2
3 Df.hist(column='Picnic', grid=False, edgecolor='black')
```

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f91d01f7cd0>]],
      dtype=object)
```



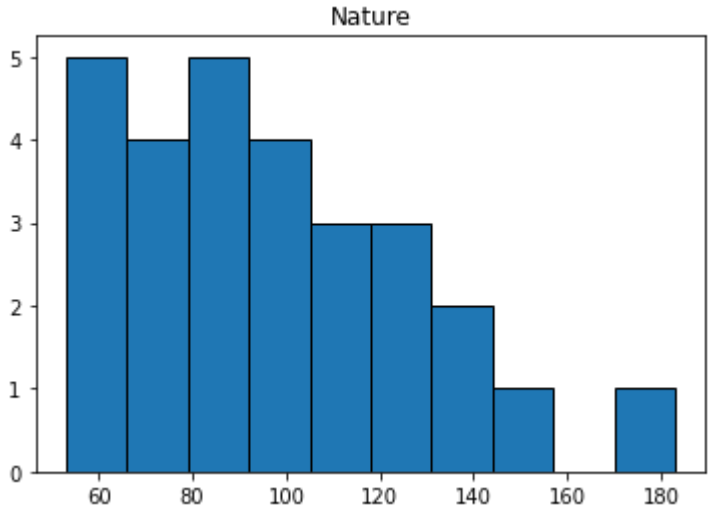

```
1 import matplotlib.pyplot as plt
2
3 Df.hist(column='Religious', grid=False, edgecolor='black')
```

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f91d01f76d0>]],
      dtype=object)
```



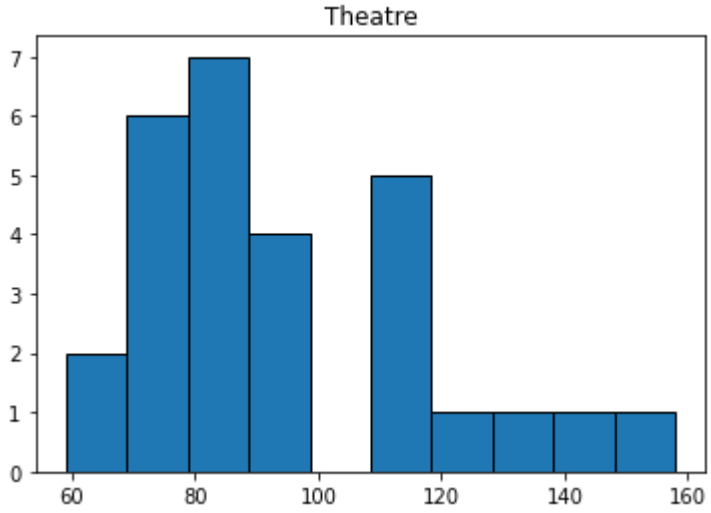

```
1 import matplotlib.pyplot as plt
2
3 Df.hist(column='Nature', grid=False, edgecolor='black')
```

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f91d00c9ad0>]],
      dtype=object)
```



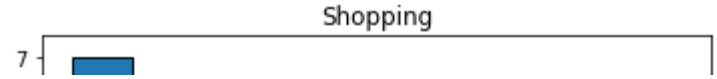

```
1 import matplotlib.pyplot as plt
2
3 Df.hist(column='Theatre', grid=False, edgecolor='black')
```

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f91d0445b90>]],
      dtype=object)
```




```
1 import matplotlib.pyplot as plt
2
3 Df.hist(column='Shopping', grid=False, edgecolor='black')
```

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f91e17ee850>]],
      dtype=object)
```

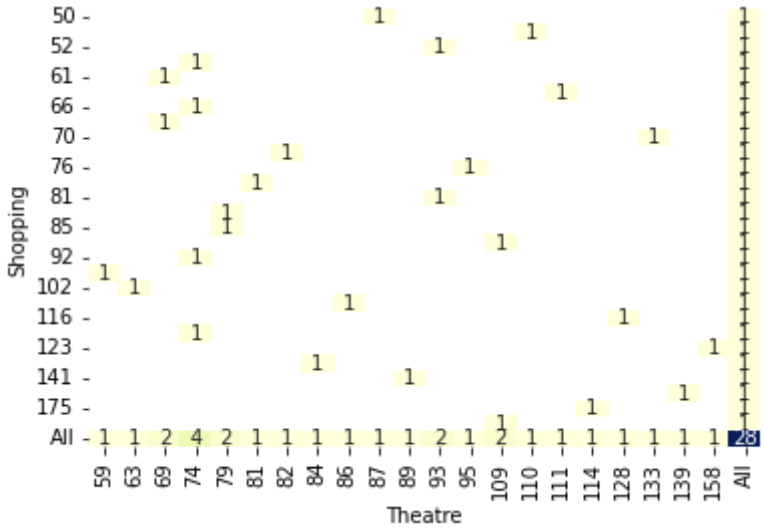


10. Bi-variable analysis: determine the interaction between variables by building visualization



```
1 import pandas as pd
2 import seaborn as sns
3 pd.crosstab(    Df.Shopping, Df.Theatre, margins=True,     values=Df.Sports, aggfunc=p
4 sns.heatmap(pd.crosstab(   Df.Shopping, Df.Theatre, margins=True,     values=Df.Sports
5             cmap="YlGnBu", annot=True, cbar=False)
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91d34de050>
```



In the above plot we can see the relation between 2 attributes shopping and theatre on the basis of sports. This helps describe if there are all three are present, any 2 are present or only 1 of the them exists

11. Scatter plot The simplest bivariate plot is the lowly scatter plot. A simple scatter plot simply maps each variable of interest to a point in two-dimensional space.

```
1 Df[Df['Shopping'] < 150].plot.scatter(x='Shopping', y='Theatre')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91cff2c150>
```



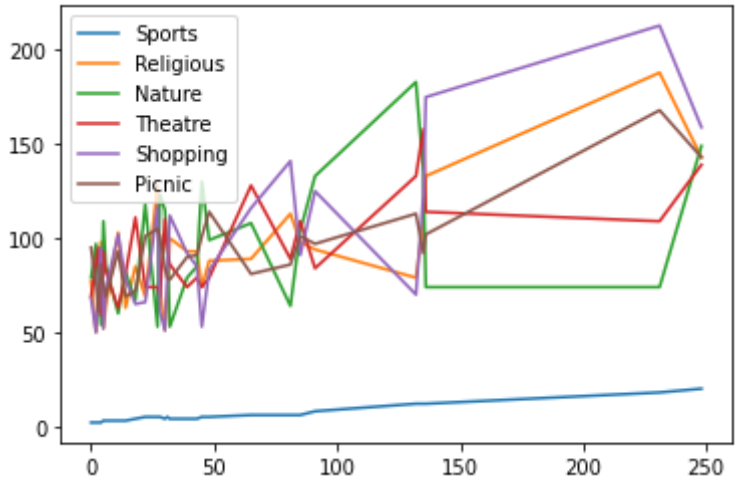
12. LINE PLOT

The line in this chart takes up so little visual space, it's really easy and effective to overplot multiple lines on the same chart.



```
1 Df.plot.line()
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91cff15210>
```



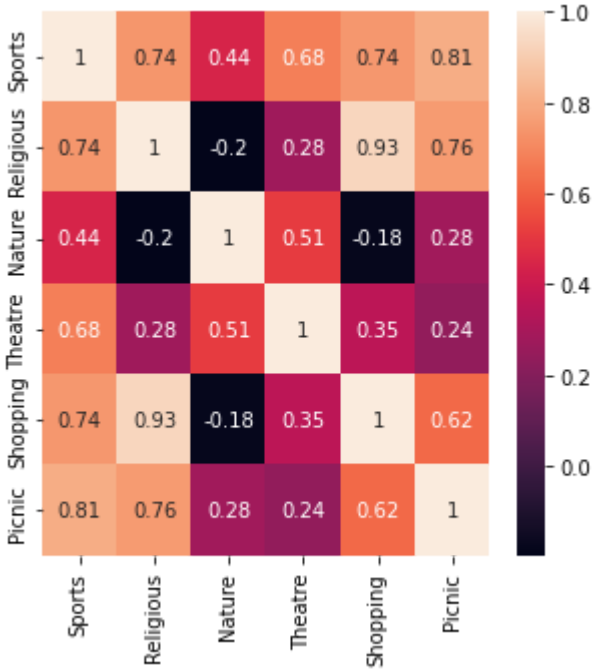
13. HEAT MAP

A heat map is a two-dimensional representation of data in which values are represented by colors. A simple heat map provides an immediate visual summary of information. More elaborate heat maps allow the viewer to understand complex data sets.

```
1 import numpy as np
2 from google.colab import drive
3 import numpy as np
4 import matplotlib.pyplot as plt
5 import pandas as pd
6 import seaborn as sns
```

```
1 correlation_matrix = Df.corr()
2 # annot = True to print the values inside the square
3 plt.subplots(figsize=(5,5))
4 sns.heatmap(data=correlation_matrix, annot=True)
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f91cfdbf250>
```



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